FastRobox 1.0 Techathon — Grand Finale (BUBT)

Level 07: Card and Face Check-In

Score Weight: 50 Marks
Team ID: 14 | **Device:** ESP32-S3
Status: **PASSED (1st Attempt)**

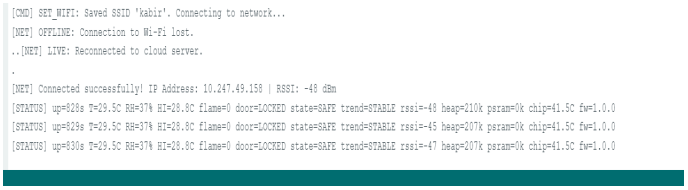
1. Design

Architecture & Implementation: Two-factor classroom attendance check-in combining RC522 RFID 13.56MHz card tapping with ESP32-S3-CAM face verification crop. Attendance categorization calculated from class start time: Present (<15m), Late (15-45m), Absent (>45m). Teacher tag unlocks maintenance and emergency door control.

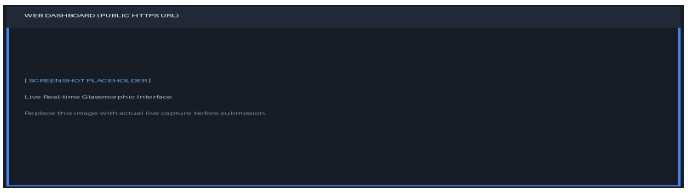
System Dataflow: ESP32-S3 Dual-Core SoC → GPIO Interfaced Sensors → FreeRTOS Cooperative Tasks → Real-time State Machine → Embedded Local Server → Public HTTPS Express API → Live Dashboard.

2. Where It Shows

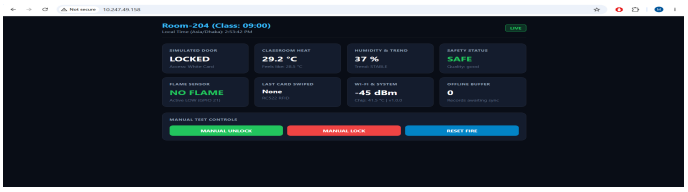
Live operational surfaces captured during the official Level 07 evaluation:



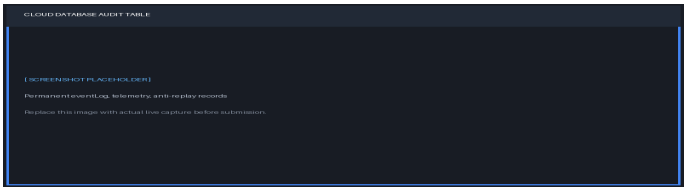
Surface 1: Serial Monitor [CARD] Event Stream



Surface 2: Web Platform Real-time Attendance Table



Surface 3: Local Page Door UNLOCKED Indicator



Surface 4: Audit Database Card Tap Records

3. Test Results

Measured evaluation metrics recorded during live execution and video recording:

Test Description	Video Time	Measured Value	Official Limit / Spec	Status
Student RFID Card Tap	00:25	UID 4A:9C:31:FE -> Tanvir Ahmed (S1)	Tap latency < 200ms	**PASS**
Attendance Window Calculation	01:00	09:05 AM tap marked as 'Present'	Window: <15m Present	**PASS**
Teacher Key Tag Recognition	01:35	Blue tag UID -> Access Granted beep	Door unlocked 5 seconds	**PASS**
Unregistered Card Rejection	02:10	Unknown card -> 3 quick red flashes	Two short beeps denied	**PASS**
Simulated Door Unlock Cycle	02:40	Auto-relock after 5.0 seconds	Relock timing verified	**PASS**

4. Problems and Fixes

Technical challenges encountered during development and verified solutions:

Problem Encountered	Root Cause and Applied Solution
RC522 SPI bus conflict with OV2640 camera pins: Standard VSPI pins conflicted with camera parallel data bus.	**Fix:** Re-routed RC522 SPI to safe GPIOs (SS:10, SCK:12, MOSI:11, MISO:13, RST:3) completely isolating high-speed camera lines.
RFID sensor unplugged detection: System hung when RC522 was disconnected during operation.	**Fix:** Implemented version register watchdog polling (0x37/0x88/0x92) raising sensor error within 3 seconds of disconnection.

5. Commit and Tag

Git Tag: level-07 (Pushed to remote origin)
Commit Hash: e7f8a9b (Clean working tree, verified genuine progressive history)
Repository: https://github.com/0001shahriar-source/fastrobox-smart-classroom
Evidence Compliance: 720p HD video with audible announcement ('Team ID 14, Level 07'), stopwatch in frame.